

Arnav Pratap

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Summary

Final-year CSE undergraduate (CGPA 9.00) and two-time IIT Patna research intern (NLP/LLM systems), with hands-on experience in full-stack web development, AWS, and algorithms. Seeking a software engineering role to build and ship dependable products.

Education

Vellore Institute of Technology, Vellore

B.Tech in Computer Science and Engineering

2023 – 2027

CGPA: 9.00 / 10

St. Karen's Secondary School

Senior Secondary (CBSE), Class XII

2020 – 2022

86%

Experience

Indian Institute of Technology, Patna (IIT Patna)

Research Intern — Multi-Agent LLM Systems

Hybrid

June 2026 – July 2026

- Architected a cost-aware **Multi-Agent Debate (MAD)** pipeline with **Llama 3 (8B)** for Hindi-Hinglish hate-speech detection, using same-model prosecutor, defender, critic, and judge roles.
- Engineered efficiency modules — Debate Necessity Predictor, Contrastive Precedent Injection, and Intra-Debate Entropy Early Exit — that skip unnecessary debates and reuse precedents, cutting token cost by up to **74.28%**.
- Raised accuracy from a 66.7% zero-shot baseline to **77.22%**, approaching the **79.8%** full-debate ceiling.

Indian Institute of Technology, Patna (IIT Patna)

Research Intern — NLP & Retrieval-Augmented Generation

Hybrid

June 2025 – August 2025

- Designed **CAD-RAG** (Context-Aware Dynamic Rationale Generation) to detect implicit, evolving hate speech beyond static classifiers, improving macro-F1 by **~15–18%** over Logistic Regression and BiLSTM baselines.
- Built a knowledge base (evolving slur lexicon, entity-level knowledge graphs, historical incident data) with a multi-query retrieval pipeline, cutting false negatives on emerging slurs by **~22%**.
- Validated on **45k+ labeled samples** across lexical, temporal, and relational signals for zero-day detection.

Projects

Mess Feedback System | Campus Platform | GitHub

HTML, CSS, JavaScript

- Deployed a campus-wide feedback system across multiple hostel blocks; collected data from **500+ students** and automated analytics, cutting admin reporting time from days to minutes.

StockSense | Stock Maintenance System | GitHub

Next.js 14, Prisma, PostgreSQL, NextAuth.js, shadcn/ui

- Developed a role-based stock-management dashboard with NextAuth.js authentication, Recharts analytics, and a Bloomberg-inspired dark UI on the Next.js 14 App Router.
- Designed a full Prisma schema with middleware-enforced RBAC and real-time inventory tracking at sub-second query performance.

StudyMate | Study-Material Organizer | GitHub

React, Node.js, AWS Textract, Comprehend, S3, DynamoDB

- Implemented a document organizer that auto-extracts keywords and topics from PDFs using **AWS Textract** (text extraction) and **AWS Comprehend** (NLP), with a React 19 + TypeScript frontend and a Node.js/Express API.
- Integrated **AWS S3** for secure PDF storage via presigned URLs and **DynamoDB** for metadata, enabling instant search across documents by keyword, topic, or filename.

Achievements & Certifications

- **Code with Cisco 2026 — Finalist:** selected for the national finals of Cisco's CSR hackathon (Bangalore).
- **Amazon ML Summer School 2026:** ranked among the top **3,000** candidates selected nationwide.
- **Competitive Programming:** solved **300+** Data Structures & Algorithms problems on LeetCode.
- **AWS Academy Cloud Architecting:** aligned to the AWS Certified Solutions Architect – Associate (SAA-C03) exam.

Technical Skills

- **Languages:** C++, Python, Java, JavaScript / TypeScript, SQL
- **Backend:** Node.js, Express.js, Next.js 14, Prisma ORM, REST APIs, WebSockets, JWT, NextAuth.js
- **Frontend & Databases:** React.js, shadcn/ui, Tailwind CSS, Recharts | PostgreSQL, MySQL, MongoDB, Redis, DynamoDB
- **Cloud & Tools:** AWS (Lambda, S3, EC2, DynamoDB, Textract, Comprehend, IAM, VPC), Git, Postman, Linux
- **Core CS:** Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, OOP, System Design
- **ML / NLP:** RAG, Multi-Agent Systems, Hugging Face Transformers, LangChain, LLM APIs, Prompt Engineering

Leadership & Extracurriculars

- **Junior Core Head, Alpha Bio Cell (ABC Club):** led technical execution for the club's flagship annual event.
- Competed in multiple **48+ hour hackathons**, shipping functional full-stack prototypes end-to-end.